

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461153.9105 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.019
Transit depth (Rp/Rs)^2	1.76 +/- 0.49 [%]
Orbital Inclination (inc)	87.17 +/- 0.49
Airmass coefficient 1 (a1)	1.099 +/- 0.011
Airmass coefficient 2 (a2)	-0.093 +/- 0.068
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	#3 - [551, 190]
Optimal Aperture	3.49
Optimal Annulus	12.18
Transit Duration (day)	0.08 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.133 ± 0.019 vs 0.1356 (deviation 0.13σ)
Duration fit vs published	0.08 d vs 0.1129 d (deviation 2.01σ)
Mid-time O-C	2.1 min
Depth SNR	6.8
Residual scatter	1.03 %

Light curve

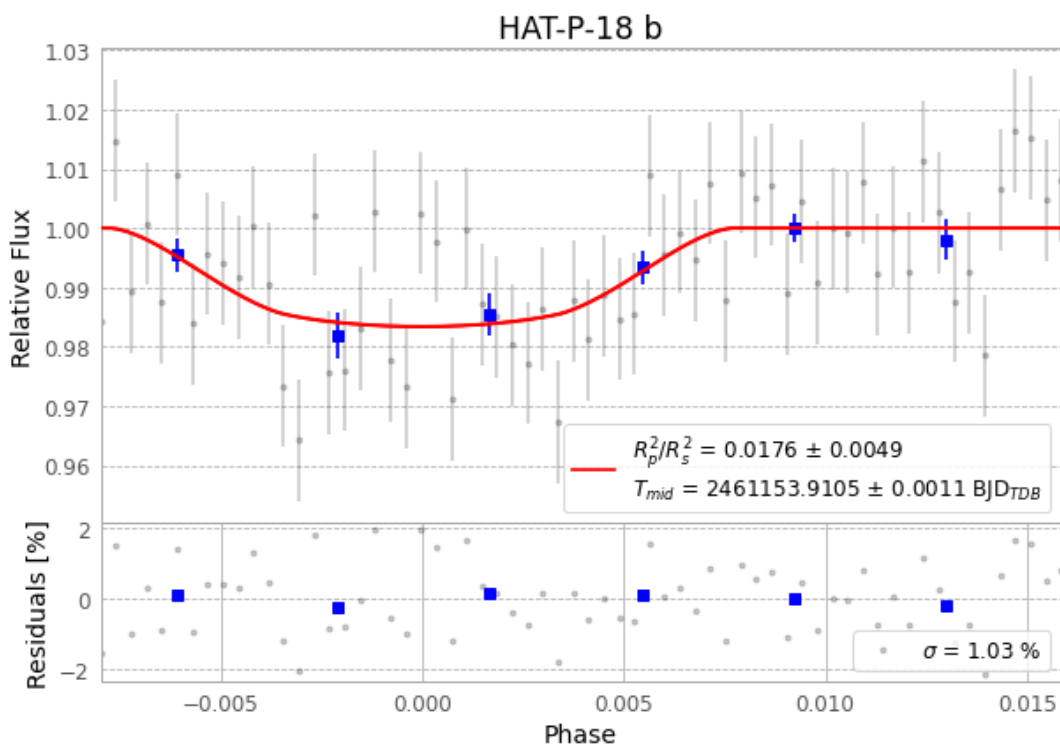


Figure 1 — FinalLightCurve_HAT-P-18 b_2026-04-23.png (SHA-256 f080a16b18051289...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 232], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 047dbff17947b967...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), HATP-18260423084217.FITS → HATP-18260423115115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-260423.log (97c2c5d69c6e...), FinalLightCurve_HAT-P-18 b_2026-04-23.png (f080a16b1805...), FinalLightCurve_HAT-P-18 b_2026-04-23.csv (41ebf8ae4a6c...)

Compute resources

Wall time 201.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 01:12Z.